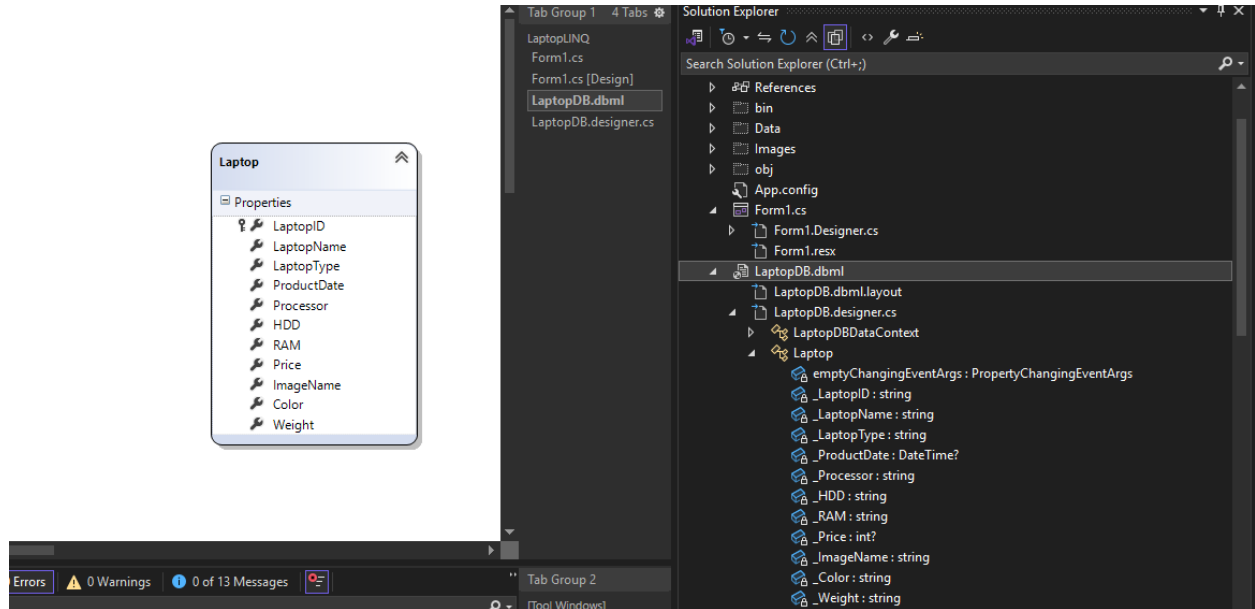


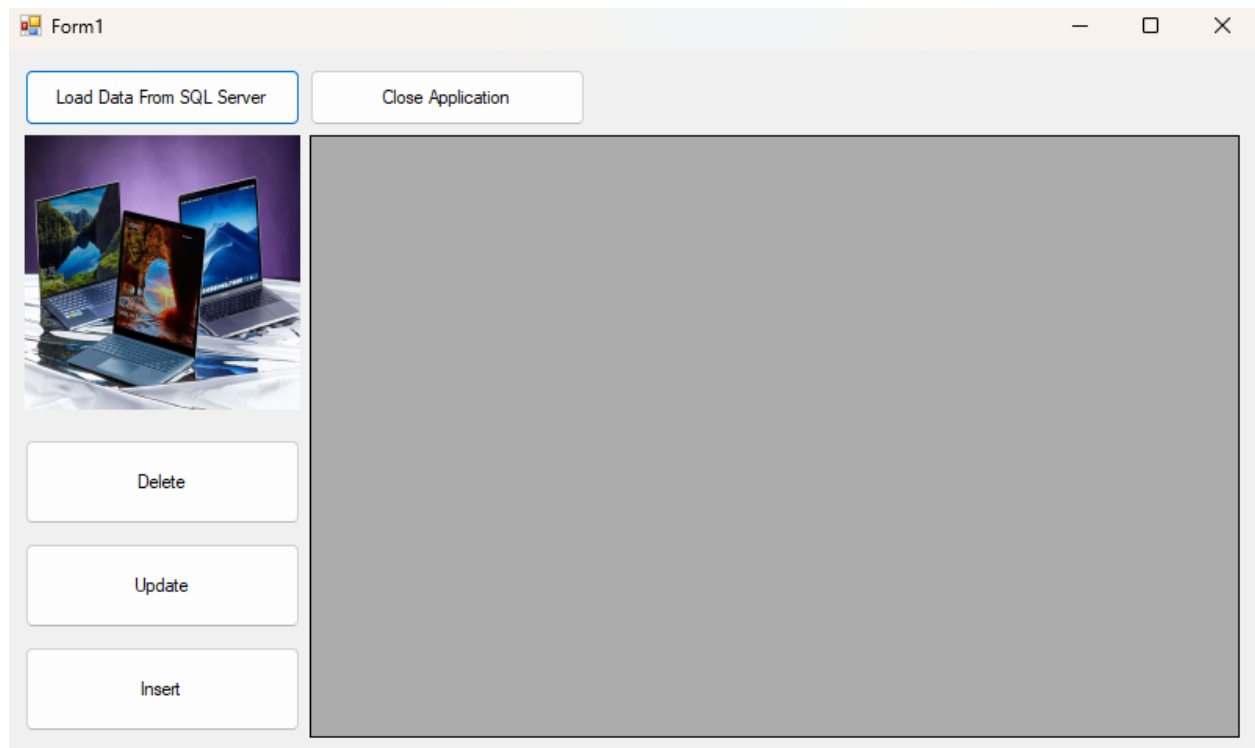
Bài Thực Hành Tuần 6 (LTNet)

1. Tạo ứng dụng quản lý LapTop đã làm dưới hình thức dùng LINQ

a/ Tạo Project và Add LINQ cơ sở dữ liệu LaptopDB



b/ Viết ứng dụng quản lý Laptop với các chức năng cơ bản



Form1.cs

```
2  using System;
3  using System.Collections.Generic;
4  using System.ComponentModel;
5  using System.Data;
6  using System.Drawing;
7  using System.Linq;
8  using System.Text;
9  using System.Threading.Tasks;
10 using System.Windows.Forms;
11 using System.IO;
12
13 namespace LaptopLINQ
14 {
15     4 references
16     public partial class Form1 : Form
17     {
18         static string ProjectPath = Directory.GetParent(Directory.GetCurrentDirectory()).Parent.FullName;
19         LaptopDBDataContext db = new LaptopDBDataContext();
20
21         1 reference
22         public Form1()
23         {
24             InitializeComponent();
25         }
26
27         0 references
28         private void Form1_Load(object sender, EventArgs e)
29         {
30             picLaptop.Image = Image.FromFile(ProjectPath + "\\Images\\laptop.jpg");
31             picLaptop.SizeMode = PictureBoxSizeMode.Zoom;
32         }
33
34         1 reference
35         private void btnCloseApp_Click(object sender, EventArgs e)
36         {
37             Application.Exit();
38         }
39
40         4 references
41         public void DisplayData()
42         {
43             dgwLaptop.DataSource = (from lt in db.Laptops select lt);
44             dgwLaptop.Columns["ImageName"].Visible = false;
45             dgwLaptop.Columns["LaptopID"].ReadOnly = true;
46         }
47
48         1 reference
49         private void btnLoadData_Click(object sender, EventArgs e)
50         {
51             DisplayData();
52         }
53
54         1 reference
55         private void dgwLaptop_SelectionChanged(object sender, EventArgs e)
56         {
57             if (dgwLaptop.Focused == false || dgwLaptop.CurrentCell == null)
58                 return;
59
60             int RowIndex = dgwLaptop.CurrentRow.Index;
61             string SelectedLaptopID = dgwLaptop.Rows[RowIndex].Cells[0].Value.ToString();
62
63             string LaptopImage = (from lp in db.Laptops
64                                   where lp.LaptopID == SelectedLaptopID
65                                   select lp.ImageName).FirstOrDefault();
66         }
67     }
68 }
```

```

59         if (LaptopImage == null)
60             return;
61
62         picLaptop.Image = Image.FromFile(ProjectPath + "\\Images\\" + LaptopImage);
63         picLaptop.SizeMode = PictureBoxSizeMode.Zoom;
64
65     }
66
67
68     1 reference
69     private void btnDelete_Click(object sender, EventArgs e)
70     {
71         int RowIndex = dgwLaptop.CurrentRow.Index;
72         string SelectedLaptopID = dgwLaptop.Rows[RowIndex].Cells[0].Value.ToString();
73
74         Laptop deletedLaptop = (from lp in db.Laptops
75                                 where lp.LaptopID == SelectedLaptopID
76                                 select lp).SingleOrDefault();
77
78         db.Laptops.DeleteOnSubmit(deletedLaptop);
79         db.SubmitChanges();
80         DisplayData();
81
82         MessageBox.Show("Laptop : " + deletedLaptop.LaptopID + " is deleted");
83     }
84
85     1 reference
86     private void btnUpdate_Click(object sender, EventArgs e)
87     {
88         if (dgwLaptop.CurrentRow == null) return;
89
90         int RowIndex = dgwLaptop.CurrentRow.Index;

```

```

91         string SelectedLaptopID = dgwLaptop.Rows[RowIndex].Cells[0].Value.ToString();
92
93         Laptop updatedLP = (from lp in db.Laptops
94                             where lp.LaptopID == SelectedLaptopID
95                             select lp).SingleOrDefault();
96
97         if (updatedLP != null)
98         {
99             updatedLP.LaptopName = dgwLaptop.Rows[RowIndex].Cells[1].Value.ToString();
100             updatedLP.LaptopType = dgwLaptop.Rows[RowIndex].Cells[2].Value.ToString();
101             updatedLP.ProductDate = DateTime.Parse(dgwLaptop.Rows[RowIndex].Cells[3].Value.ToString());
102             updatedLP.Processor = dgwLaptop.Rows[RowIndex].Cells[4].Value.ToString();
103             updatedLP.HDD = dgwLaptop.Rows[RowIndex].Cells[5].Value.ToString();
104             updatedLP.RAM = dgwLaptop.Rows[RowIndex].Cells[6].Value.ToString();
105             updatedLP.Price = Convert.ToInt32(dgwLaptop.Rows[RowIndex].Cells[7].Value.ToString());
106             updatedLP.ImageName = "laptop.jpg";
107
108             db.SubmitChanges();
109             DisplayData();
110             MessageBox.Show("Laptop: " + updatedLP.LaptopID + " is updated");
111         }
112
113     1 reference
114     private void btnAdd_Click(object sender, EventArgs e)
115     {
116         Laptop checkedLP;
117         List<Laptop> LPList = (from lp in db.Laptops
118                               where lp.LaptopID.Contains("NEW")
119                               select lp).ToList();

```

```

120     string NewID = "NEW";
121     if (LPList.Count == 0)
122         NewID = NewID + "001";
123     else
124     {
125         checkedLP = LPList.Last();
126         NewID = NewID + NextId(checkedLP.LaptopID.Substring(checkedLP.LaptopID.Length - 3));
127     }
128
129     Laptop insertedLP = new Laptop();
130     insertedLP.LaptopID = NewID;
131     insertedLP.LaptopName = "Not Assigned";
132     insertedLP.LaptopType = "Not Assigned";
133     insertedLP.ProductDate = DateTime.Now;
134     insertedLP.Processor = "Not Assigned";
135     insertedLP.HDD = "Not Assigned";
136     insertedLP.RAM = "Not Assigned";
137     insertedLP.Price = 0;
138     insertedLP.ImageName = "Laptop.jpg";
139     insertedLP.Color = "Not Assigned";
140     insertedLP.Weight = "Not Assigned";
141
142     db.Laptops.InsertOnSubmit(insertedLP);
143     db.SubmitChanges();
144     DisplayData();
145     MessageBox.Show("New Laptop is inserted! Please update data and press button Update");
146 }
147
148 1 reference
149 public string NextId(string currentId)
150 {
151     int i = 0;

```

```

151     if (int.TryParse(currentId, out i))
152     {
153         i++;
154         return (i.ToString().PadLeft(3, '0'));
155     }
156     throw new System.ArgumentException("Non-numeric string passed as argument");
157 }
158 }
159 }

```

c/ Kết quả chạy chương trình

Nhấn nút Load Data From SQL Server và chọn Laptop tùy ý

Form1

Load Data From SQL Server Close Application



Delete

Update

Insert

LaptopID	LaptopName	LaptopType	ProductDate	Processor	HDD
LP001	Dell XPS 13 Plus	Ultrabook	1/1/2026	Core i7-1360P	512GE
LP002	MacBook Air M2	Ultrabook	1/1/2026	Apple M2	256GE
LP003	Asus ROG Strix ...	Gaming	1/1/2026	Ryzen 7 6800H	1TB S
LP004	HP Spectre x360	2-in-1	1/1/2026	Core i7-1255U	1TB S
LP005	Lenovo Legion 5	Gaming	1/1/2026	Core i5-12500H	512GE
LP006	Acer Swift 5	Văn phòng	1/1/2026	Core i7-1260P	512GE
LP007	MSI Raider GE78	Gaming	1/1/2026	Core i9-13980HX	2TB S
LP008	MacBook Pro 14	Đồ họa	1/1/2026	Apple M2 Pro	512GE
LP009	LG Gram 17	Văn phòng	1/1/2026	Core i7-1360P	1TB S
LP010	Surface Laptop 5	Văn phòng	1/1/2026	Core i5-1235U	256GE
LP011	Dell Inspiron 15	Sinh viên	1/1/2026	Core i5-1135G7	512GE
LP012	Acer Nitro 5	Gaming	1/1/2026	Core i5-12500H	512GE
LP013	ThinkPad X1 Car...	Doanh nhân	1/1/2026	Core i7-1355U	1TB S
LP014	Asus ZenBook 14	Văn phòng	1/1/2026	Core i5-1240P	512GE
LP015	HP Victus 16	Gaming	1/1/2026	Ryzen 5 6600H	512GE
LP016	Lenovo Yoga 9i	2-in-1	1/1/2026	Core i7-1360P	1TB S

Form1

Load Data From SQL Server Close Application



Delete

Update

Insert

LaptopID	LaptopName	LaptopType	ProductDate	Processor	HDD
LP001	Dell XPS 13 Plus	Ultrabook	1/1/2026	Core i7-1360P	512GE
LP002	MacBook Air M2	Ultrabook	1/1/2026	Apple M2	256GE
LP003	Asus ROG Strix ...	Gaming	1/1/2026	Ryzen 7 6800H	1TB S
LP004	HP Spectre x360	2-in-1	1/1/2026	Core i7-1255U	1TB S
LP005	Lenovo Legion 5	Gaming	1/1/2026	Core i5-12500H	512GE
LP006	Acer Swift 5	Văn phòng	1/1/2026	Core i7-1260P	512GE
LP007	MSI Raider GE78	Gaming	1/1/2026	Core i9-13980HX	2TB S
LP008	MacBook Pro 14	Đồ họa	1/1/2026	Apple M2 Pro	512GE
LP009	LG Gram 17	Văn phòng	1/1/2026	Core i7-1360P	1TB S
LP010	Surface Laptop 5	Văn phòng	1/1/2026	Core i5-1235U	256GE
LP011	Dell Inspiron 15	Sinh viên	1/1/2026	Core i5-1135G7	512GE
LP012	Acer Nitro 5	Gaming	1/1/2026	Core i5-12500H	512GE
LP013	ThinkPad X1 Car...	Doanh nhân	1/1/2026	Core i7-1355U	1TB S
LP014	Asus ZenBook 14	Văn phòng	1/1/2026	Core i5-1240P	512GE
LP015	HP Victus 16	Gaming	1/1/2026	Ryzen 5 6600H	512GE
LP016	Lenovo Yoga 9i	2-in-1	1/1/2026	Core i7-1360P	1TB S

Thực hiện Update Laptop LP009 Laptop Type thành Sinh viên

Form1

Load Data From SQL Server Close Application



Delete Update Insert

LaptopID	LaptopName	LaptopType	ProductDate	Processor	HDD
LP001	Dell XPS 13 Plus	Ultrabook	1/1/2026	Core i7-1360P	512GE
LP002	MacBook Air M2	Ultrabook	1/1/2026	Apple M2	256GE
LP003	Asus ROG Strix ...	Gaming	1/1/2026	Ryzen 7 6800H	1TB S.
LP004	HP Spectre x360	2-in-1	1/1/2026	Core i7-1255U	1TB S.
LP005	Lenovo Legion 5	Gaming	1/1/2026	Core i5-12500H	512GE
LP006	Acer Swift 5	Văn phòng	1/1/2026	Core i7-1260P	512GE
LP007	MSI Raider GE78	Gaming	1/1/2026	Core i9-13980HX	2TB S.
LP008	MacBook Pro 14	Đồ họa	1/1/2026	Apple M2 Pro	512GE
LP009	LG Gram 17	Sinh viên	1/1/2026	Core i7-1360P	1TB S.
LP010	Surface Laptop 5	Văn phòng	1/1/2026	Core i5-1250U	256GE
LP011	Dell Inspiron 15	Sinh viên	1/1/2026	Core i5-1135G7	512GE
LP012	Acer Nitro 5	Gaming	1/1/2026	Core i5-12500H	512GE
LP013	ThinkPad X1 Car...	Doanh nhân	1/1/2026	Core i7-1355U	1TB S.
LP014	Asus ZenBook 14	Văn phòng	1/1/2026	Core i5-1240P	512GE
LP015	HP Victus 16	Gaming	1/1/2026	Ryzen 5 6600H	512GE
LP016	Lenovo Yoga 9i	2-in-1	1/1/2026	Core i7-1360P	1TB S.

Laptop: LP009 is updated

OK

Thực hiện Delete Laptop LP010

Form1

Load Data From SQL Server Close Application



Delete Update Insert

LaptopID	LaptopName	LaptopType	ProductDate	Processor	HDD
LP001	Dell XPS 13 Plus	Ultrabook	1/1/2026	Core i7-1360P	512GE
LP002	MacBook Air M2	Ultrabook	1/1/2026	Apple M2	256GE
LP003	Asus ROG Strix ...	Gaming	1/1/2026	Ryzen 7 6800H	1TB S.
LP004	HP Spectre x360	2-in-1	1/1/2026	Core i7-1255U	1TB S.
LP005	Lenovo Legion 5	Gaming	1/1/2026	Core i5-12500H	512GE
LP006	Acer Swift 5	Văn phòng	1/1/2026	Core i7-1260P	512GE
LP007	MSI Raider GE78	Gaming	1/1/2026	Core i9-13980HX	2TB S.
LP008	MacBook Pro 14	Đồ họa	1/1/2026	Apple M2 Pro	512GE
LP009	LG Gram 17	Sinh viên	1/1/2026	Core i7-1360P	1TB S.
LP011	Dell Inspiron 15	Sinh viên	1/1/2026	Core i5-1135G7	512GE
LP012	Acer Nitro 5	Gaming	1/1/2026	Core i5-12500H	512GE
LP013	ThinkPad X1 Car...	Doanh nhân	1/1/2026	Core i7-1355U	1TB S.
LP014	Asus ZenBook 14	Văn phòng	1/1/2026	Core i5-1240P	512GE
LP015	HP Victus 16	Gaming	1/1/2026	Ryzen 5 6600H	512GE
LP016	Lenovo Yoga 9i	2-in-1	1/1/2026	Core i7-1360P	1TB S.
LP017	Razer Blade 15	Gaming	1/1/2026	Core i7-12800H	1TB S.

Laptop :LP010 is deleted

OK

Thực hiện Insert thêm Laptop mới

Form1

Load Data From SQL Server

Close Application



Delete

Update

Insert

LaptopID	LaptopName	LaptopType	ProductDate	Processor	HDD
LP009	LG Gram 17	Sinh viên	1/1/2026	Core i7-1360P	1TB S.
LP011	Dell Inspiron 15	Sinh viên	1/1/2026	Core i5-1135G7	512GE
LP012	Acer Nitro 5	Gaming	1/1/2026	Core i5-12500H	512GE
LP013	ThinkPad X1 Car...	Doanh nhân	1/1/2026	Core i7-1355U	1TB S.
LP014	Asus ZenBook 14	Văn phòng	1/1/2026	Core i5-1240P	512GE
LP015	HP Victus 16	Gaming	1/1/2026	Ryzen 5 6600H	512GE
LP016	Lenovo Yoga 9i	2-in-1	1/1/2026	Core i7-1360P	1TB S.
LP017	Razer Blade 15	Gaming	1/1/2026	Core i7-12800H	1TB S.
LP018	Samsung Galaxy ...	Văn phòng	1/1/2026	Core i7-1360P	512GE
LP019	MSI Modern 14	Sinh viên	1/1/2026	Core i3-1115G4	256GE
LP020	Alienware x15 R2	Gaming	1/1/2026	Core i9-12900H	2TB S.
NEW001	Not Assigned	Not Assigned	1/7/2026 2:58 PM	Not Assigned	Not As
NEW002	Not Assigned	Not Assigned	1/7/2026 2:58 PM	Not Assigned	Not As
NEW003	Not Assigned	Not Assigned	1/7/2026 2:58 PM	Not Assigned	Not As
*					

2. Tạo ứng dụng quản lý Coffee hình thức dùng LINQ và áp dụng Binding Datasource

a/ Thiết kế giao diện

The screenshot shows a Windows Forms application titled 'Form1'. The form is divided into two main sections. On the left is a data entry form with the following fields: 'Drink ID:', 'Drink Name:', 'Category:', 'Price:', 'Description:', 'Image Name:', 'Date Added:' (with a date picker set to 'Wednesday, January 7, 2026'), 'Unit:', 'In Stock:', 'Is Best Seller:' (with a checkbox), and 'Discount:'. On the right is a data grid with the following columns: 'DrinkID', 'DrinkName', 'Category', 'Price', and 'Description'. The grid currently contains one row with a '*' in the 'DrinkID' column. At the bottom of the application, there is a taskbar with five icons and labels: 'bindingNavigator1', 'coffeeDBDataSet', 'coffeeBindingSource', 'coffeeTableAdapter', and 'tableAdapterManager'.

b/ Form1.cs

```
1  using System;
2  using System.Collections.Generic;
3  using System.ComponentModel;
4  using System.Data;
5  using System.Drawing;
6  using System.Linq;
7  using System.Text;
8  using System.Threading.Tasks;
9  using System.Windows.Forms;
10 using System.IO;
11
12 namespace CoffeeLINQ
13 {
14     4 references
15     public partial class Form1 : Form
16     {
17         1 reference
18         public Form1()
19         {
20             InitializeComponent();
21
22         1 reference
23         private void Form1_Load(object sender, EventArgs e)
24         {
25             this.coffeeTableAdapter.Fill(this.coffeeDBDataSet.Coffee);
26         }
27     }
28 }
```

```

26 private void coffeeBindingNavigatorSaveItem_Click(object sender, EventArgs e)
27 {
28     try
29     {
30         this.Validate();
31         this.coffeeBindingSource.EndEdit();
32         this.tableAdapterManager.UpdateAll(this.coffeeDBDataSet);
33         MessageBox.Show("Data saved successfully!", "Notification");
34     }
35     catch (Exception ex)
36     {
37         MessageBox.Show("Error: " + ex.Message, "Error");
38     }
39 }
40

```

1 reference

```

41 private void coffeeBindingSource_CurrentChanged(object sender, EventArgs e)
42 {
43     if (coffeeBindingSource.Current == null) return;
44
45     var currentRow = (DataRowView)coffeeBindingSource.Current;
46     string imageName = currentRow["ImageName"].ToString();
47
48     string folderPath = Application.StartupPath + "\\..\\..\\Images\\";
49     string fullPath = folderPath + imageName;
50
51     if (!string.IsNullOrEmpty(imageName) && System.IO.File.Exists(fullPath))
52     {
53         picCoffee.ImageLocation = fullPath;
54     }
55     else
56     {
57         string defaultImgPath = folderPath + "cf_denda.jpg";
58
59         if (System.IO.File.Exists(defaultImgPath))
60         {
61             picCoffee.ImageLocation = defaultImgPath;
62         }
63         else
64         {
65             picCoffee.Image = null;
66         }
67     }
68 }
69
70

```

```

70 1 reference
71 private void coffeeBindingSource_AddingNew(object sender, AddingNewEventArgs e)
72 {
73     DataView view = (DataView)coffeeBindingSource.List;
74     DataRowView newRow = view.AddNew();
75
76     string finalID = "CF001";
77
78     if (view.Count > 1)
79     {
80         var prevRow = view[view.Count - 2];
81         string lastID = prevRow["DrinkID"].ToString();
82
83         if (!string.IsNullOrEmpty(lastID) && lastID.Length >= 3)
84         {
85             try
86             {
87                 string numberPart = lastID.Substring(2);
88                 int nextNumber = int.Parse(numberPart) + 1;
89                 finalID = "CF" + nextNumber.ToString("D3");
90             }
91             catch
92             {
93                 finalID = "CF001";
94             }
95         }
96
97         newRow["DrinkID"] = finalID;
98         newRow["DrinkName"] = "New Item";
99         newRow["DateAdded"] = DateTime.Now;

```

```

100         newRow["Price"] = 0;
101         newRow["InStock"] = 0;
102         newRow["ImageName"] = "";
103         newRow["IsBestSeller"] = false;
104         newRow["Discount"] = 0;
105
106         e.NewObject = newRow;
107         coffeeBindingSource.MoveLast();
108     }
109 }
110

```

c/ Kết quả chạy chương trình:

Form1

2 of 10



Drink ID: CF002

Drink Name: Iced Milk Coffee

Category: Coffee

Price: 29000

Description: Traditional Vietnamese style

Image Name: cf_suada.jpg

Date Added: Sunday, January 15, 2023

Unit: Cup

In Stock: 85

Is Best Seller: ☒

Discount: 0

DrinkID	DrinkName	Category	Price	Description
CF001	Iced Black Coffee	Coffee	25000	Strong, pure taste
CF002	Iced Milk Coffee	Coffee	29000	Traditional Vietna...
CF003	Cappuccino	Machine Coffee	45000	Frothy and creamy
CF004	Peach Lemongra...	Fruit Tea	42000	Fresh and cooling
CF005	Matcha Ice Blen...	Ice Blended	55000	Japanese Green ...
CF006	Avocado Smoothie	Smoothie	49000	Smooth and buttery
CF007	Orange Juice	Juice	40000	100% pure fresh ...
CF008	Yogurt with Pearls	Snack	35000	Ha Long style yo...
CF009	Tiramisu	Cake	45000	Rich cocoa flavor
CF010	Salted Egg Croiss...	Savory Cake	35000	Crispy layers
*				

Thực hiện Add 2 lần trên BindingNavigator

Form1

12 of 12



Drink ID: CF012

Drink Name: New Item

Category:

Price: 0

Description:

Image Name:

Date Added: Thursday, January 8, 2026

Unit:

In Stock: 0

Is Best Seller: ☐

Discount: 0

DrinkID	DrinkName	Category	Price	Description
CF001	Iced Black Coffee	Coffee	25000	Strong, pure taste
CF002	Iced Milk Coffee	Coffee	29000	Traditional Vietna...
CF003	Cappuccino	Machine Coffee	45000	Frothy and creamy
CF004	Peach Lemongra...	Fruit Tea	42000	Fresh and cooling
CF005	Matcha Ice Blen...	Ice Blended	55000	Japanese Green ...
CF006	Avocado Smoothie	Smoothie	49000	Smooth and buttery
CF007	Orange Juice	Juice	40000	100% pure fresh ...
CF008	Yogurt with Pearls	Snack	35000	Ha Long style yo...
CF009	Tiramisu	Cake	45000	Rich cocoa flavor
CF010	Salted Egg Croiss...	Savory Cake	35000	Crispy layers
CF011	New Item		0	
CF012	New Item		0	
*				

Thực hiện lưu dữ liệu mới vào Database bằng nút Save trên BindingNavigator

Form1

12 of 12

+

✗

📄

📁

🖨

✂

📋

📌

🔍



Drink ID: CF012

Drink Name: New Item

Category:

Price: 0

Description:

Image Name:

Date Added: Thursday, January 8, 2026

Unit:

In Stock: 0

Is Best Seller: ☐

Discount: 0

DrinkID	DrinkName	Category	Price	Description
CF001	Iced Black Coffee	Coffee	25000	Strong, pure taste
CF002	Iced Milk Coffee	Coffee	29000	Traditional Vietna...
CF003	Cappuccino	Machine Coffee	45000	Frothy and creamy
CF004	Peach Lemongra...	Fruit Tea	42000	Fresh and cooling
CF005	Matcha Ice Blen...	Ice Blended	55000	Japanese Green ...
CF006	Avocado S		000	Smooth and buttery
CF007	Orange Jui		000	100% pure fresh ...
CF008	Yogurt with		000	Ha Long style yo...
CF009	Tiramisu		000	Rich cocoa flavor
CF010	Salted Egg		000	Crispy layers
CF011	New Item			
▶ CF012	New Item		0	
*				

Notification

✕

Data saved successfully!

OK

[illegible]